T5-E1-IPEX Module Datasheet

Version: 20260715

Online Version

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T5-E1-IPEX is a highly integrated single-antenna single-band 2.4 GHz Wi-Fi 6 (IEEE 802.11b/g/n/ax) and Bluetooth 5.4 Low Energy (LE) combo IoT module. Featuring a multi-peripheral packaging and ultra-low-power chip, the T5-E1-IPEX module provides a highly integrated, efficient, secure, and lowest-power environment for IP cameras, HMI applications, smart locks, and other advanced IoT applications.

> *i*
>
> The T5-E1-IPEX module is not pin-to-pin compatible with the T5-E1 module.

1. Features

1.1. CPU and on-chip memory

- Built-in with Tuya’s custom T5QN88 chip, Armv8-M Star (M33F) processor, with a clock frequency of up to 480 MHz
- 8 MB flash memory
- 16 MB PSRAM
- 640 KB shared SRAM
- 64 KB ROM

1.2. Wi-Fi

- IEEE 802.11b/g/n/ax compliant
- 20 MHz and 40 MHz channel bandwidth
- Support downlink multi-user multiple input, multiple output (DL MU-MIMO)
- Support orthogonal frequency-division multiple access (OFDMA)
- Support target wake time (TWT)
- Integrated Bluetooth and Wi-Fi coexistence (packet traffic arbitration, PTA)
- Transmit (TX) power up to +20 dBm
- Receive (RX) sensitivity of −99 dBm

1.3. Bluetooth

- Support Bluetooth LE 5.4 standard
- Support Bluetooth LE 1 Mbit/s and long-range mode (125 Kbit/s and 500 Kbit/s)
- TX power of +6 dBm
- RX sensitivity of −97 dBm

1.4. Peripherals

- 56x GPIOs
- 2x SPIs and 2x QSPIs
- 3x UART interfaces: 1 channel has hardware flow control and supports flash download

- 1x smart card controller
- 1x SDIO interface, 2x I2C interfaces, and 1x CAN controller with CAN FD
- 1x high-speed (HS) USB 2.0 interface
- 1x display controller supporting RGB565 and 8080 interfaces
- 1x segment LCD controller for up to 4 × 32 segments
- 1x 8-bit CIS DVP interface
- 1x 720p H.264 video encoder
- 1x Ethernet MAC interface
- 5x 32-bit PWM channels
- 3x I2S interfaces
- 2x audio ADCs and 1x audio DAC
- 12-bit AUX ADC, up to 11 channels
- 1x touch sensor, up to 16 touch sensing I/Os

1.5. Operating conditions

- Operating/supply voltage range: 2.5 V to 3.6 V

> *i*
>
> In the T5-E1-IPEX module, VBAT and VIO are directly connected, meaning VBAT = VIO. To support peripherals such as SD cards and DVP cameras, the recommended VBAT operating voltage range is 3.0 V to 3.6 V.

- Operating temperature range: −40°C to 85°C

1.6. Module certification

- Wi-Fi & Bluetooth: FCC and CE
- Environmental certification: REACH/RoHS
- Bluetooth certification: BQB

1.7. Reliability test

- High and low temperature, high temperature and high humidity, high temperature, and cold start tests
- Salt spray and electrostatic discharge tests

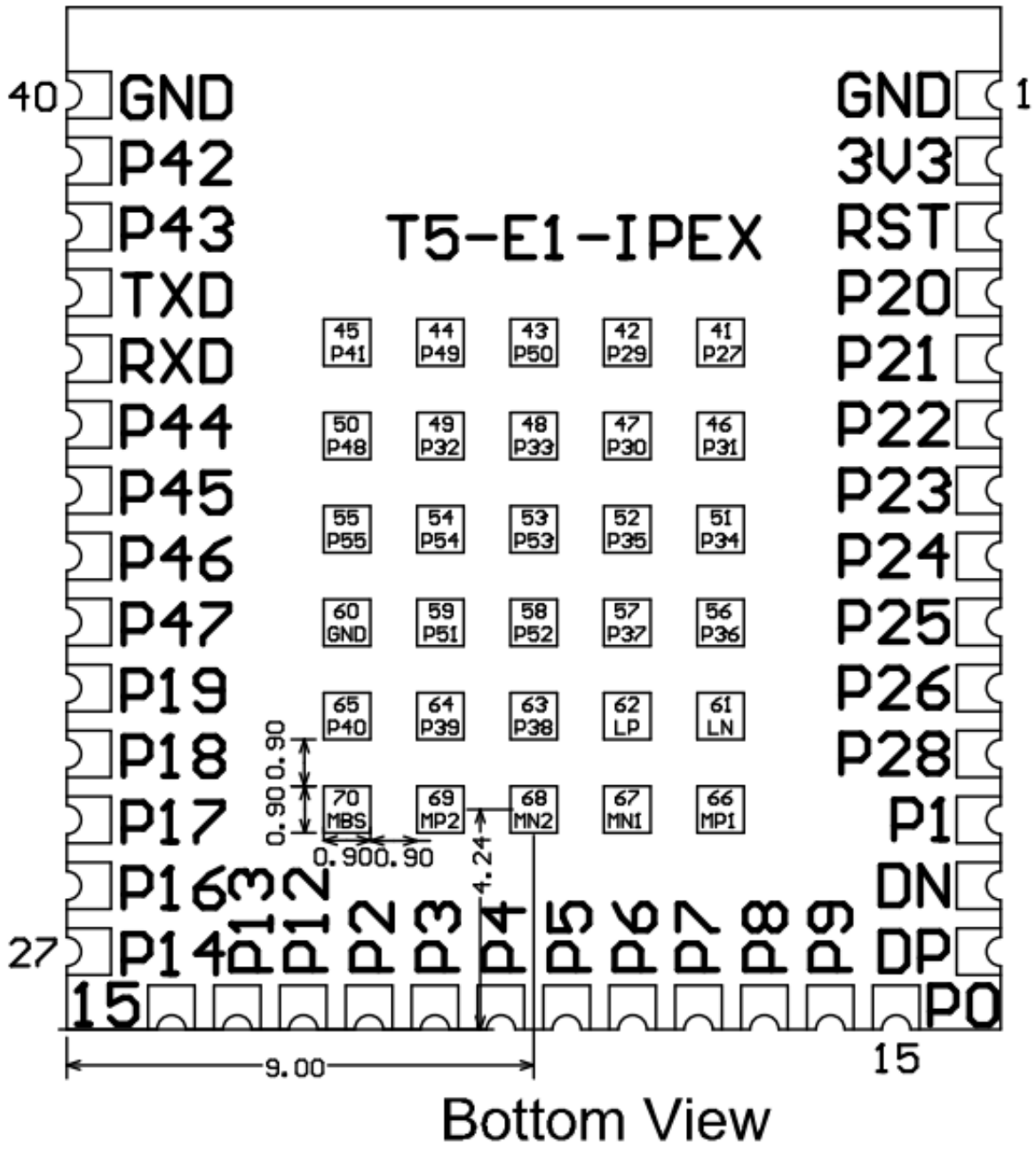
2. Scope of applications

- Human-machine interaction (HMI) application
- Home appliance
- Air conditioner
- Refrigerator
- Thermostat
- Dishwasher
- Robot vacuum
- Smart socket
- Smart lighting
- Industrial wireless control
- Baby monitor
- IP camera and smart lock
- Smart bus

3. Pin definition

3.1. Pin layout

The pin layout diagram shows the approximate location of the pins on the module. For the actual layout drawn to scale, refer to the module dimensional drawings. The top view looks like this:



3.2. Pin definition

The module has a total of 70 pins. For a detailed description, refer to the table below.

No.	Name	I/O type	Description
1	GND	P	Ground pin
2	3V3	P	Power pin
3	RST	I	The low-level reset pin, active low, and pulled up internally
4	P20	I/O	• GPIO20 • I2C0_SCL • SWCLK • RGB_R6 • I8080_D9 • SEG10
5	P21	I/O	• I2C0_SDA • SWDIO • ADC6 • RGB_R5 • I8080_D8 • SEG9
6	P22	I/O	• CLK26M • ADC5 • QSPI0_SCK • RGB_R4 • I8080_CSX • SEG8

No.	Name	I/O type	Description
7	P23	I/O	• ADC3 • QSPI0_CS • RGB_R3 • I8080_RESET • SEG7
8	P24	I/O	• LPO_CLK • PWMG0_PWM4 • ADC2 • QSPI0_IO0 • RGB_G7 • I8080_RSX • SEG6
9	P25	I/O	• IRDA • ADC1 • QSPI0_IO1 • RGB_G6 • I8080_WRX • SEG5
10	P26	I/O	• QSPI0_IO2 • RGB_G5 • I8080_RDX • SEG4
11	P28	I/O	• I2S_MCLK • ADC4 • TOUCH2 • CLK_AUXS_CIS • SEG18

No.	Name	I/O type	Description
12	P1	I/O	• UART1_RX • I2C1_SDA • SWDIO • SC_CLK • ADC13 • LIN_RXD
13	DN	I/O	USB D-
14	DP	I/O	USB D+
15	P0	I/O	• UART1_TX • I2C1_SCL • SWCLK • SC_IO • ADC12 • LIN_TXD
16	P9	I/O	• I2S0_DOUT • DMIC_DAT • 32K_XI
17	P8	I/O	• I2S0_DIN • DMIC_CLK • ADC10 • 32K_XO
18	P7	I/O	• I2S0_SYNC • QSPI1_IO3
19	P6	I/O	• I2S0_SCK • QSPI1_IO2
20	P5	I/O	• SPI1_MISO • SDIO_DATA1 • COM7 • QSPI1_IO1 • SEG31

No.	Name	I/O type	Description
21	P4	I/O	• SPI1_MOSI • SDIO_DATA0 • COM6 • QSPI1_IO0 • SEG30
22	P3	I/O	• SPI1_CSN • SDIO_CMD • SC_VCC • QSPI1_CS
23	P2	I/O	• SPI1_SCK • SDIO_CLK • SC_RSTN • LIN_SLEEP • QSPI1_SCK
24	P12	I/O	• UART0_RTS • TOUCH0 • ADC14
25	P13	I/O	• UART0_CTS • TOUCH1 • ADC15
26	P15	I/O	• SDIO_CMD • SPI0_CSN • I2C1_SDA • RGB_DISP • I8080_D14 • SEG15
27	P14	I/O	• SDIO_CLK • SPI0_SCK • I2C1_SCL • RGB_DCLK • I8080_D15 • SEG16

No.	Name	I/O type	Description
28	P16	I/O	• SDIO_DATA0 • SPI0_MOSI • RGB_DE • I8080_D13 • SEG14
29	P17	I/O	• SDIO_DATA1 • SPI0_MISO • RGB_HSYNC • I8080_D12 • SEG13
30	P18	I/O	• SDIO_DATA2 • PWMG0_PWM0 • RGB_VSYNC • I8080_D11 • SEG12
31	P19	I/O	• SDIO_DATA3 • RGB_R7 • I8080_D10 • SEG11
32	P47	I/O	• SPI0_MISO • ENET_MDC • TOUCH15 • RGB_B3 • I8080_D0 • COM0 • I2S2_DOUT

No.	Name	I/O type	Description
33	P46	I/O	• CAN_STBY • SPI0_MOSI • ENET_PHY_INT • TOUCH14 • RGB_B4 • I8080_D1 • COM1 • I2S2_DIN
34	P45	I/O	• CAN_RX • SPI0_CSN • RGB_B5 • I8080_D2 • COM2 • I2S2_SYNC
35	P44	I/O	• CAN_TX • SPI0_SCK • RGB_B6 • I8080_D3 • COM3 • I2S2_SCK
36	RXD	I/O	< • P10 • DL_UART_RX • SDIO_DATA2 • CLK_AUXS_CIS • UART0_RX • Flash firmware and authorize modules

No.	Name	I/O type	Description
37	TXD	I/O	• P11 • DL_UART_TX • UART0_TX • SDIO_DATA3 • Flash firmware and authorize modules
38	P43	I/O	• I2C1_SDA • I2S1_DOUT • SC_VCC • RGB_B7 • I8080_D4 • SEG0
39	P42	I/O	• I2C1_SCL • I2S1_DIN • LIN_SLEEP • SC_RSTN • RGB_G2 • I8080_D5 • SEG1
40	GND	P	Ground pin
41	P27	I/O	• CIS_MCLK • CLK_AUXS_CIS • ENET_PHY_INT • QSPI0_IO3 • SEG17
42	P29	I/O	• CIS_PCLK • ENET_MDC • TOUCH3 • SEG19
43	P50	I/O	• ENET_RXD1 • RGB_R0

No.	Name	I/O type	Description
44	P49	I/O	• ENET_RXD0 • RGB_R1 • I8080_D17
45	P41	I/O	• UART2_TX • I2S1_SYNC • LIN_TXD • SC_IO • RGB_G3 • I8080_D6 • SEG2
46	P31	I/O	• CIS_VSYNC • UART2_TX • LIN_TXD • TOUCH5 • SC_IO • SEG21
47	P30	I/O	• CIS_HSYNC • UART2_RX • LIN_RXD • TOUCH4 • SC_CLK • SEG20
48	P33	I/O	• CIS_PXD1 • ENET_RXD0 • TOUCH7 • SPI0_SCK • SEG23

No.	Name	I/O type	Description
49	P32	I/O	• CIS_PXD0 • PWMG1_PWM0 • ENET_MDIO • TOUCH6 • SC_RSTN • SEG22
50	P48	I/O	• ENET_MDIO • RGB_R2 • I8080_D16
51	P34	I/O	• CIS_PXD2 • PWMG1_PWM2 • ENET_RXD1 • TOUCH8 • SPI0_CSN • SEG24
52	P35	I/O	• CIS_PXD3 • ENET_RXDV • TOUCH9 • SPI0_MOSI • SEG25
53	P53	I/O	• ENET_TXD1 • RGB_B2
54	P54	I/O	• ENET_TXEN • RGB_B1
55	P55	I/O	• ENET_REF_CLK • RGB_B0
56	P36	I/O	• CIS_PXD4 • PWMG1_PWM4 • ENET_TXD0 • TOUCH10 • SPI0_MISO • SEG26

No.	Name	I/O type	Description
57	P37	I/O	• CIS_PXD5 • ENET_TXD1 • TOUCH11 • SEG27
58	P52	I/O	• ENET_TXD0 • RGB_G0
59	P51	I/O	• ENET_RXDV • RGB_G1
60	GND	P	Ground pin
61	LN	AO	Audio left channel negative output
62	LP	AO	Audio left channel positive output
63	P38	I/O	• CIS_PXD6 • I2C1_SCL • ENET_TXEN • TOUCH12 • COM4 • SEG28
64	P39	I/O	• CIS_PXD7 • I2C1_SDA • ENET_REF_CLK • TOUCH13 • COM5 • SEG29
65	P40	I/O	• UART2_RX • I2S1_SCK • LIN_RXD • SC_CLK • RGB_G4 • I8080_D7 • SEG3

No.	Name	I/O type	Description
66	MP1	AI	Microphone 1 positive input
67	MN1	AI	Microphone 1 negative input
68	MN2	AI	Microphone 2 negative input
69	MP2	AI	Microphone 2 positive input
70	MBS	AO	Microphone bias output

- Due to the presence of power-up glitches, pins P0, P4, P5, P8, P10, P11, P20, P21, P22 and P23 are not recommended for applications sensitive to voltage fluctuations during power-up, such as LED drivers or motor drivers.
- The maximum input voltage of the ADC pin is 1.1V. It is recommended to use MΩ-level resistors for the external voltage divider, with a 100nF filter capacitor placed close to the ADC pin and connected to ground.
- P indicates the power pin.
- I/O indicates the input and output pin.
- AI indicates the analog signal input pin.
- AO indicates the analog signal output pin.

4. Electrical parameters

4.1. Absolute electrical parameters

Parameter	Description	Min value	Max value	Unit
Ts	Storage temperature	−55	125	°C
VBAT	Supply voltage	−0.3	3.6	V
Electrostatic discharge voltage (human body model)	TAMB −25°C	−4	4	kV
Electrostatic discharge voltage (machine model)	TAMB −25°C	−200	200	V

4.2. Normal operating conditions

Parameter	Description	Min value	Typical value	Max value	Unit
Ta	Operating temperature	−40	-	85	°C
VBAT	Supply voltage	2.5	3.3	3.6	V
VOL	I/O low-level output	VSS	-	VSS + 0.3	V
VOH	I/O high-level output	VBAT − 0.3	-	VBAT	V

Parameter	Description	Min value	Typical value	Max value	Unit
Imax	I/O drive current	-	6	20	mA
θ	Supply voltage slope	100	-	-	mV/ms

4.3. Radio frequency (RF) power

Operating status	Mode	Rate	Transmit/ Receive power	Average value	Peak (Typical) value	Unit
Transmit	11B	11 Mbit/s	+17 dBm	-	322	mA
Transmit	11G	54 Mbit/s	+15 dBm	-	275	mA
Transmit	11N	HT20 MCS7	+14 dBm	-	265	mA
Transmit	11AX	HE20 MCS7	+14 dBm	-	264	mA
Receive	11B	11 Mbit/s	Continuous reception	-	23	mA
Receive	11G	54 Mbit/s	Continuous reception	-	23	mA
Receive	11N	HT20 MCS7	Continuous reception	-	23	mA
Receive	11AX	HE20 MCS7	Continuous reception	-	23	mA

In order to test the transmitting operating current, the module is in the state of sending packets at 100% duty cycle.

4.4. Operating current

Working mode	Status (Ta = 25°C)	Average value	Max (Typical) value	Unit
Bluetooth pairing	The module is pairing over Bluetooth. The network status indicator blinks slowly.	58	280	mA
Access point (AP) pairing	The module is pairing in AP mode. The network status indicator blinks slowly.	60	270	mA
Connected	The module is connected to the cloud. The network status indicator is steady on.	32	270	mA

Working mode	Status (Ta = 25°C)	Average value	Max (Typical) value	Unit
Weakly connected	The connection between the module and the access point is intermittent. The network status indicator is steady on.	120	320	mA
Disconnected	The module is disconnected from the cloud. The network status indicator is steady off.	37	270	mA
Module disabled	The module’s clock enable (CEN) pin is pulled down.	133	-	μA

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The above data module is powered by a 3.3 V regulated power supply with firmware v1.0.9. There are slight differences in test data under different test environment conditions.

5. RF parameters

5.1. Basic RF features

Parameter	Description
Operating frequency	2.412 to 2.484 GHz
Wi-Fi standard	IEEE 802.11b/g/n/ax (channels 1–14)
Data transmission rate	• IEEE 802.11b: 1, 2, 5.5, and 11 Mbit/s • IEEE 802.11g: 6, 9, 12, 18, 24, 36, 48, and 54 Mbit/s • IEEE 802.11n: HT20 MCS0-7 and HT40 MCS0-7 • IEEE 802.11ax: HE20 MCS0-7 and HE40 MCS0-7
Antenna type	External antenna

5.2. Wi-Fi transmitter (TX) performance

Parameter	Min value	Typical value	Max value	Unit
RF average output power, 802.11b CCK mode, 11 Mit/s	15	17	19	dBm
RF average output power, 802.11g OFDM mode, 54 Mbit/s	13	15	17	dBm

Parameter	Min value	Typical value	Max value	Unit
RF average output power, 802.11n OFDM mode, HT20 MCS7	12	14	16	dBm
RF average output power, 802.11n OFDM mode, HT40 MCS7	11	13	15	dBm
RF average output power, 802.11ax OFDMA mode, HE20 MCS7	12	14	16	dBm
RF average output power, 802.11ax OFDMA mode, HE40 MCS7	11	13	15	dBm
Frequency error	−20	-	20	ppm

5.3. Wi-Fi receiver (RX) performance

Parameter	Min value	Typical value	Max value	Unit
PER < 8%, RX sensitivity, 802.11b DSSS mode, 11 Mbit/s	-	−88	-	dBm

Parameter	Min value	Typical value	Max value	Unit
PER < 10%, RX sensitivity, 802.11g OFDM mode, 54 Mbit/s	-	−74	-	dBm
PER < 10%, RX sensitivity, 802.11n OFDM mode, HT40 MCS7	-	−73	-	dBm
PER < 10%, RX sensitivity, 802.11ax OFDMA mode, HE40 MCS7	-	−73	-	dBm
PER < 10%, RX sensitivity, Bluetooth LE, 1 Mbit/s	-	−97	-	dBm

5.4. Bluetooth TX performance

Parameter	Min value	Typical value	Max value	Unit
Operating frequency	2402	-	2480	MHz
Transmission rate over the air	-	-	-	Mbit/s
Transmission power	−20	6	15	dBm
Frequency error	−150	-	150	KHz

5.5. Bluetooth RX performance

Parameter	Min value	Typical value	Max value	Unit
RX sensitivity	-	−97	-	dBm
Max RF signal input	−10	-	-	dBm
Intermodulation	-	-	−23	dBm
Adjacent-channel rejection ratio	-	10	-	dB

6. Antenna information

6.1. Antenna type

The T5-E1-IPEX module uses an external antenna (3rd generation IPEX antenna connector).

6.2. Antenna interference reduction

The distance between the antenna and other mental components should be at least 15 mm to provide the best Wi-Fi performance.

7. Development guide

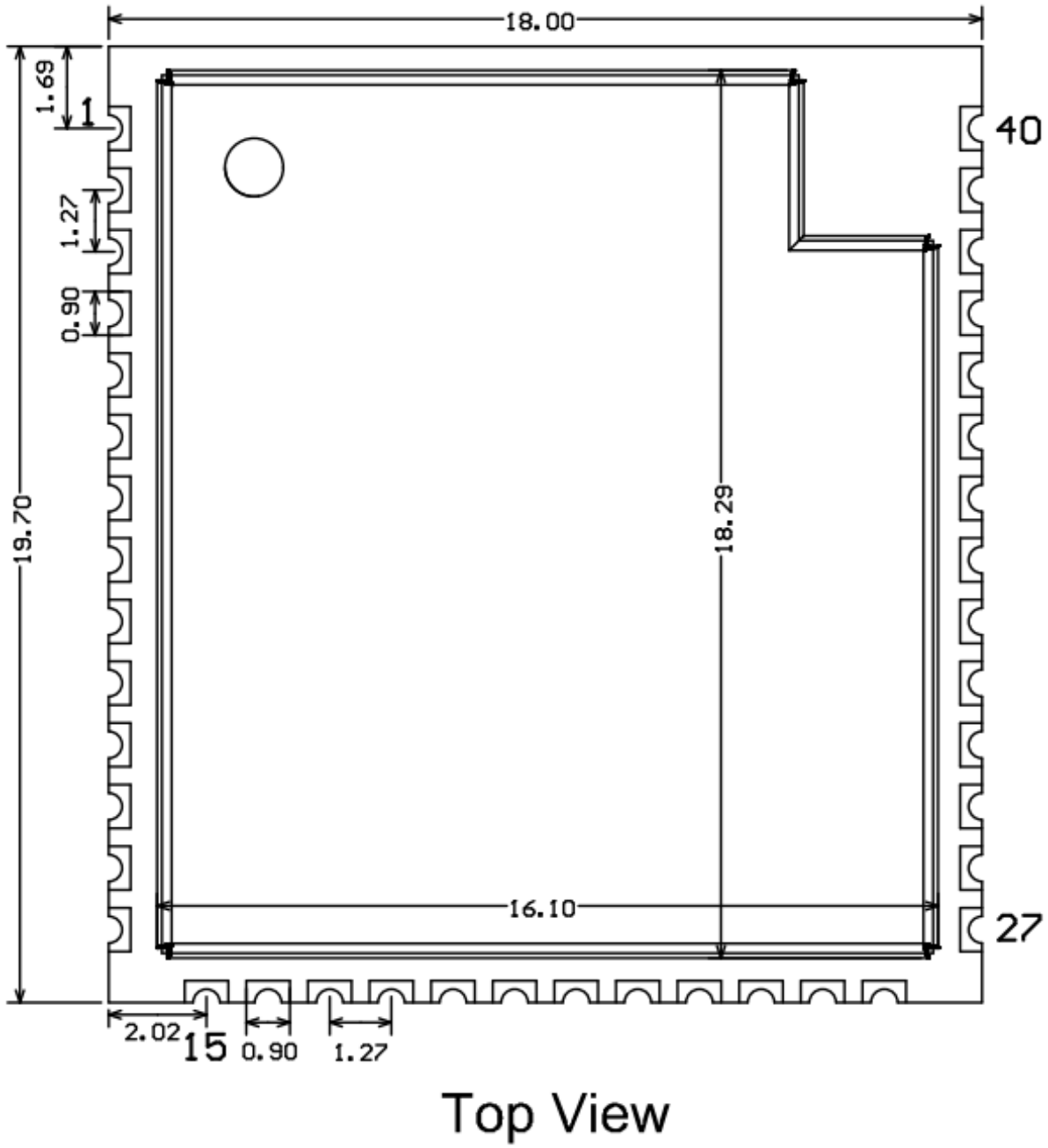
7.1. MCU integration

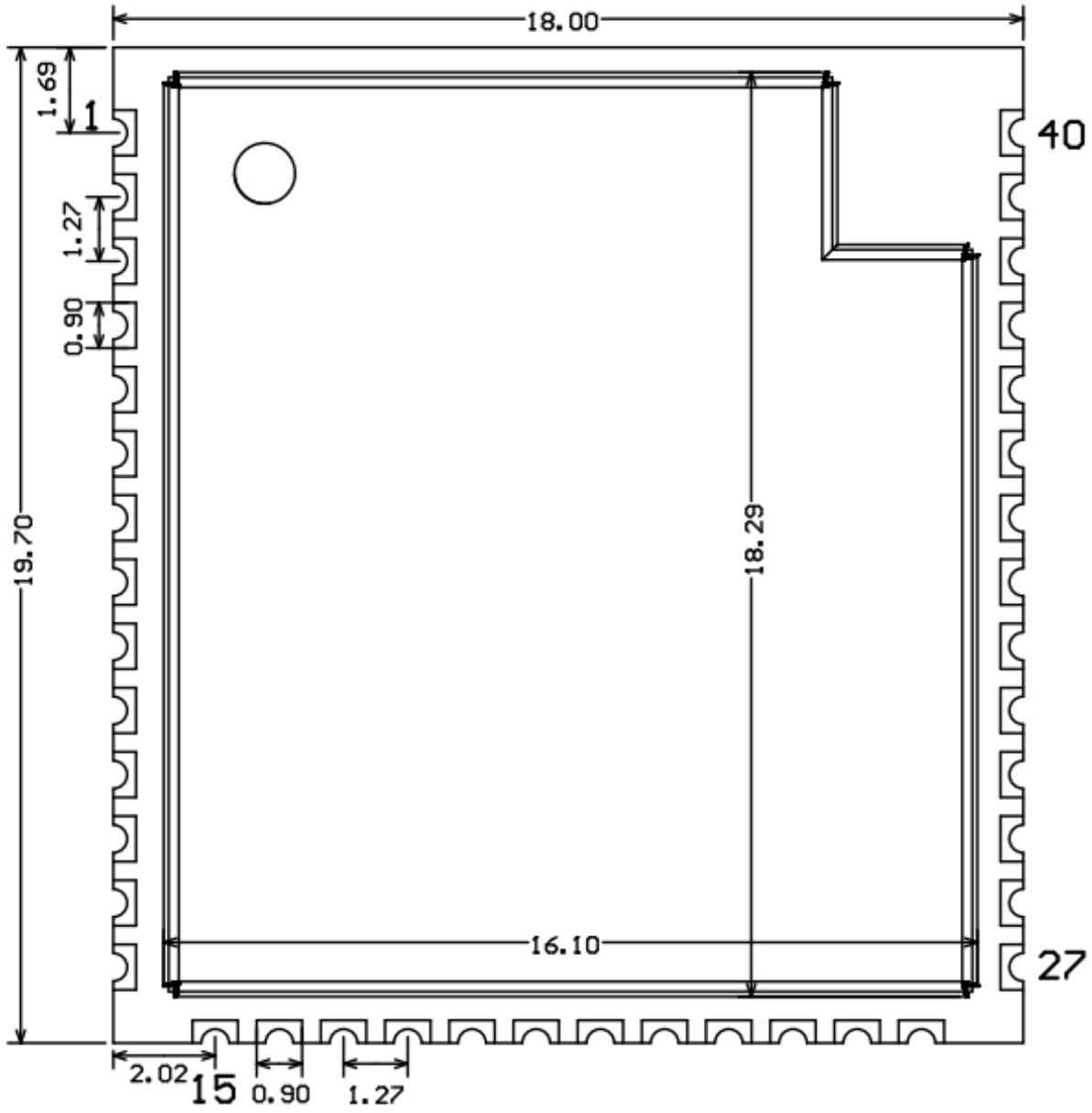
For more information about MCU general integration, see Hardware Design of CBx Series Modules .

8. Packing and production instructions

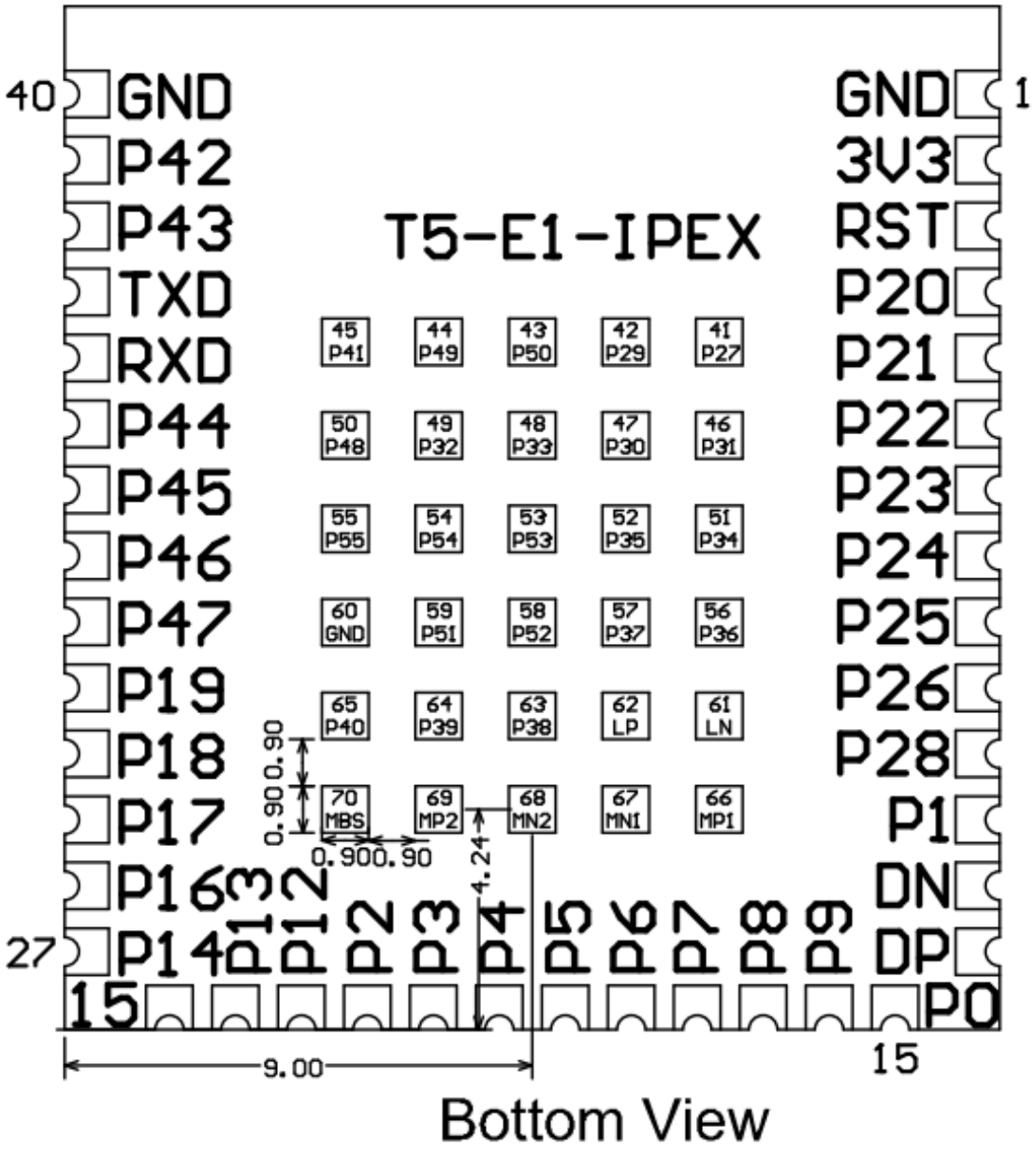
8.1. Mechanical dimensions

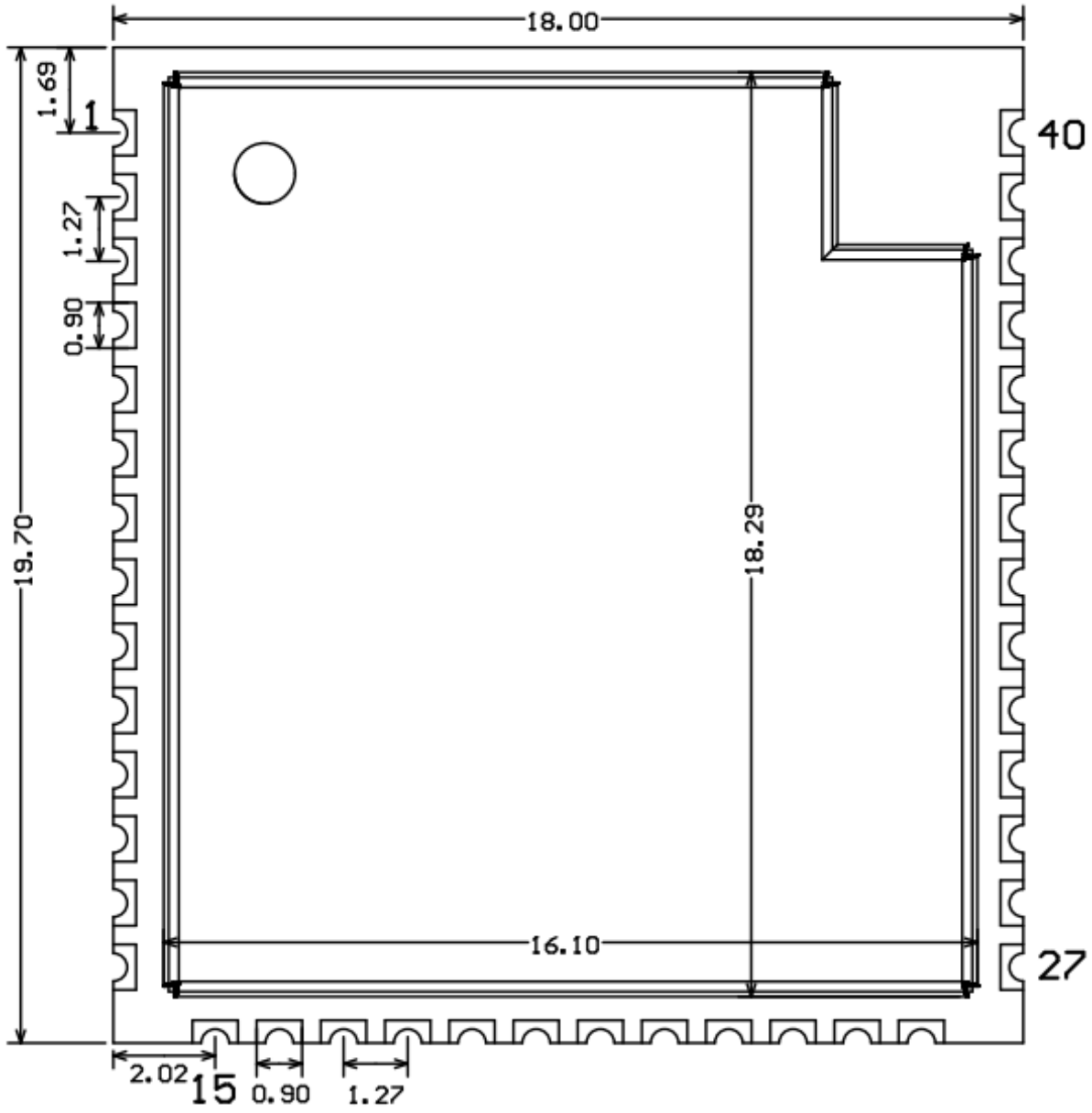
The T5-E1-IPEX dimensions are 18.00±0.35 mm (W) × 19.70±0.35 mm (L) × 2.8±0.15 mm (H).



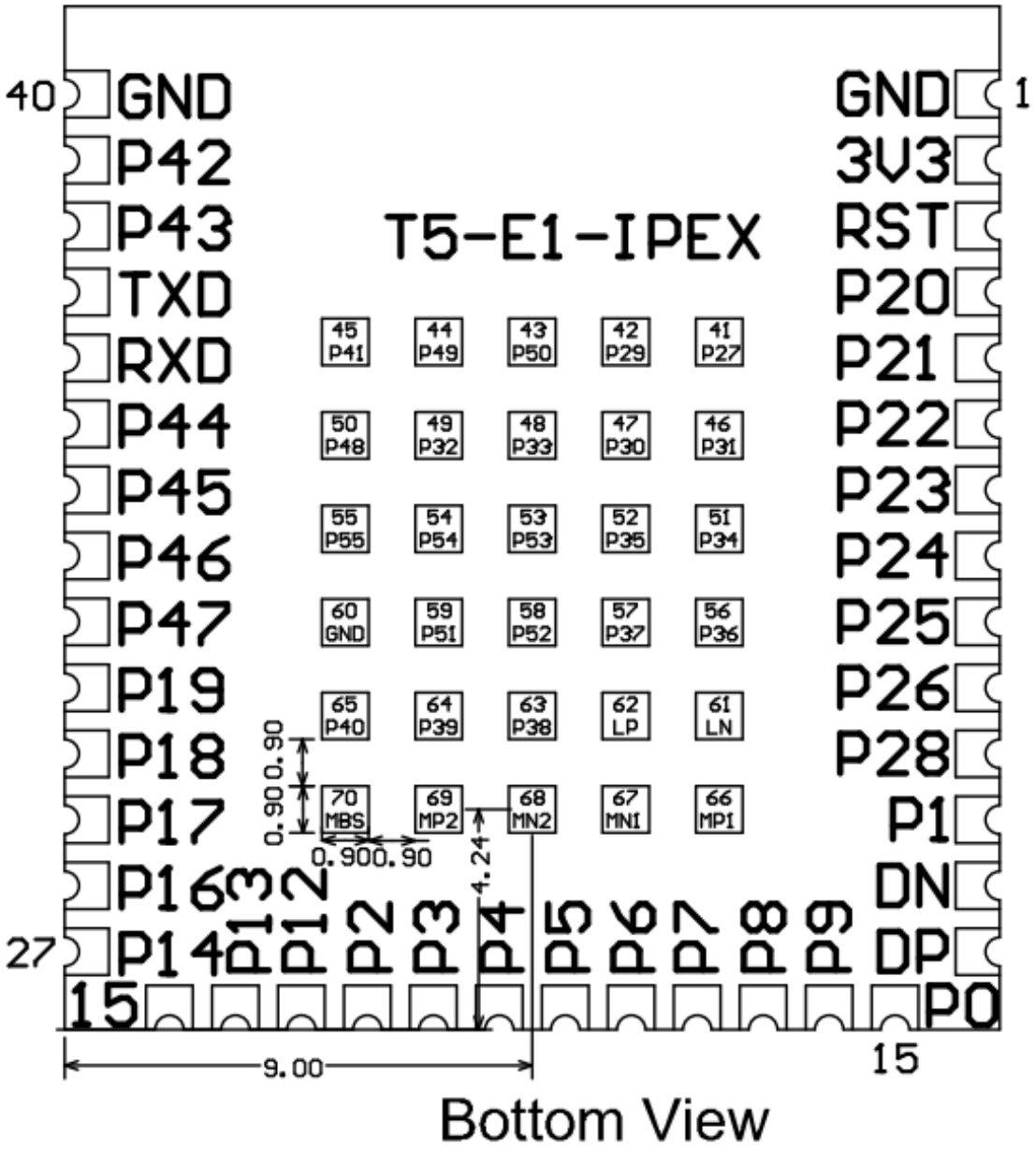


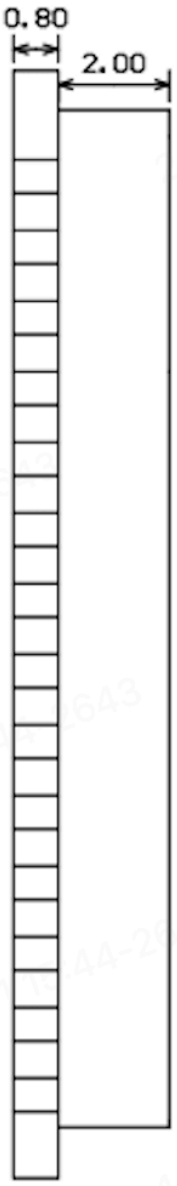
Top View


T5-E1-IPEX
40
GND
P42
P43
TXD
RXD
P44
P45
P46
P47
P19
P18
P17
P16
P14
27
15
GND
1
3V3
RST
P20
P21
P22
P23
P24
P25
P26
P28
P1
DN
DP
P0
15
45 P41
44 P49
43 P50
42 P29
41 P27
50 P48
49 P32
48 P33
47 P30
46 P31
55 P55
54 P54
53 P53
52 P35
51 P34
60 GND
59 P51
58 P52
57 P37
56 P36
65 P40
64 P39
63 P38
62 LP
61 LN
70 MBS
69 MP2
68 MN2
67 MNI
66 MPI
P13
P12
P2
P3
P4
P5
P6
P7
P8
P9
0.90
0.90
0.90
0.90
4.24
9.00
Bottom View



Top View


T5-E1-IPEX
40
GND
P42
P43
TXD
RXD
P44
P45
P46
P47
P19
P18
P17
P16
P14
27
1
GND
3V3
RST
P20
P21
P22
P23
P24
P25
P26
P28
P1
DN
DP
P0
15
45 P41
44 P49
43 P50
42 P29
41 P27
50 P48
49 P32
48 P33
47 P30
46 P31
55 P55
54 P54
53 P53
52 P35
51 P34
60 GND
59 P51
58 P52
57 P37
56 P36
65 P40
64 P39
63 P38
62 LP
61 LN
70 MBS
69 MP2
68 MN2
67 MNI
66 MPI
0.90
0.90
0.90
0.90
4.24
9.00
Bottom View



Unit： mm

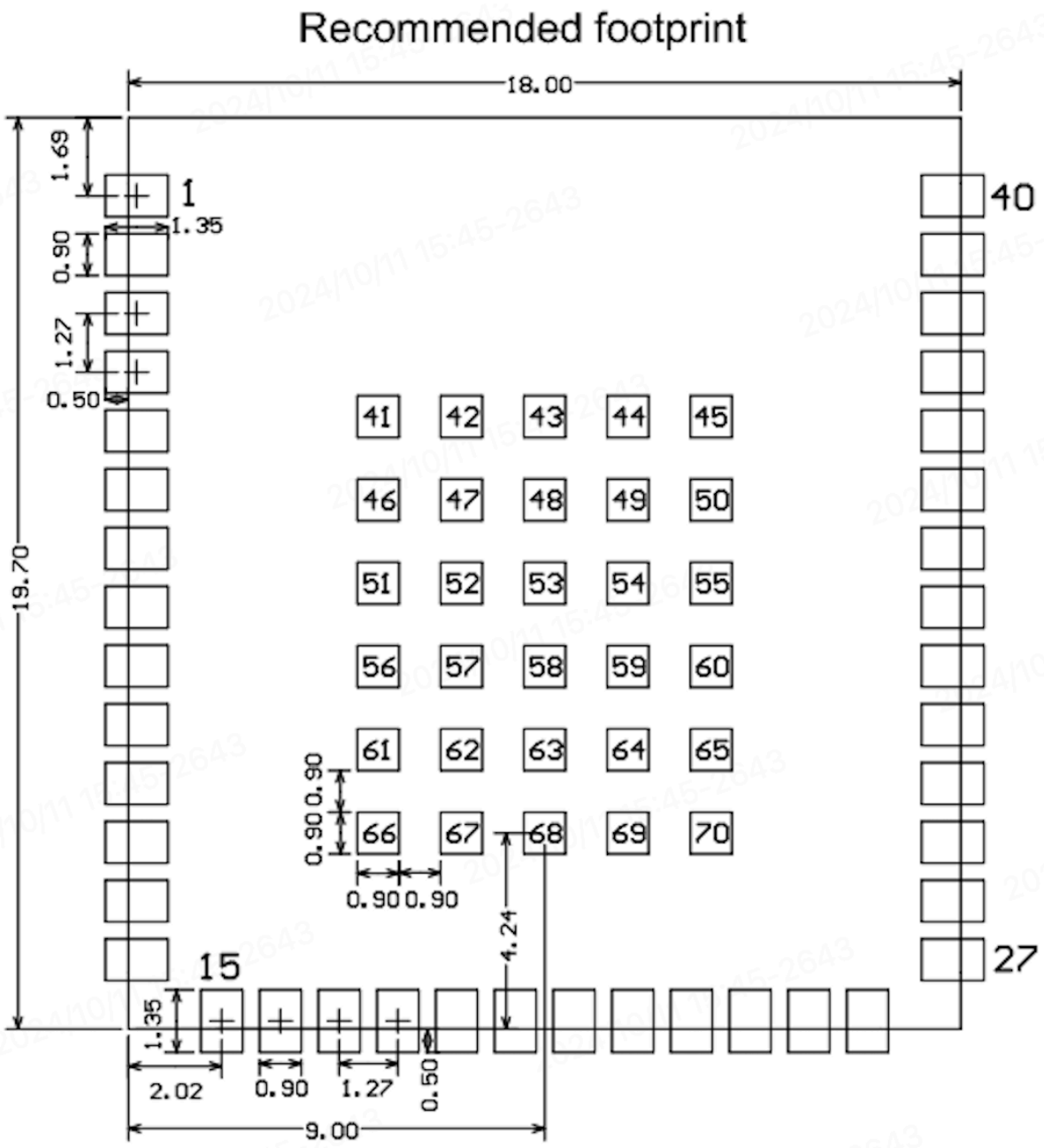
Module form factor tolerance: ± 0.35mm

PCB thickness tolerance: ±0.1mm

Shield cover height tolerance: ±0.05mm

Side View

8.2. Recommended footprint



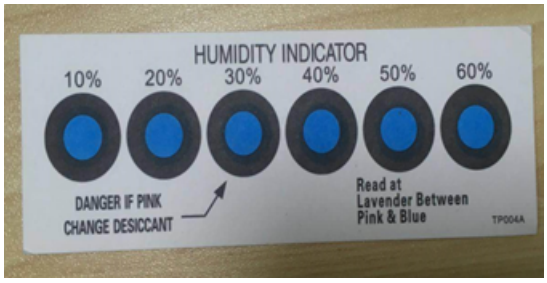
8.3. Production instructions

1. Package the module with the SMT if Tuya’s module is designed to be SMT-packaged. After being unpacked, the module must be soldered within 24 hours. Otherwise, it needs to be put into a drying cupboard with a relative humidity level no greater than 10%, or pack the module in vacuum again. Then, record the packing time and duration of exposure. The total exposure time cannot exceed 168 hours.

- Instruments or devices required for the SMT process:
- Surface mount system
- SPI
- Reflow soldering machine
- Thermal profiler
- Automated optical inspection (AOI) equipment
- Instruments or devices required for the baking process:
- Cabinet oven
- Electrostatic discharge (ESD) protection and heat-resistant trays
- ESD protection and heat-resistant gloves

2. A delivered module must meet the following storage requirements:

- The moisture-proof bag must be placed in an environment where the temperature is below 40°C and the relative humidity is lower than 90%.
- The shelf life of a dry-packaged product is 12 months from the date when the product is packaged and sealed.
- A humidity indication card (HIC) is put in the sealed package.



3. The module needs to be baked in the following cases:

- The vacuum packaging bag is damaged before unpacking.
- After unpacking, no HIC is found in the packaging bag.
- After unpacking, the HIC indicates a humidity level of 10% or higher. In this case, the circle turns pink on the HIC.
- The total exposure time has lasted for over 168 hours since unpacking.
- More than 12 months have passed since the first sealing of the bag.

4. The baking parameter settings are described below:

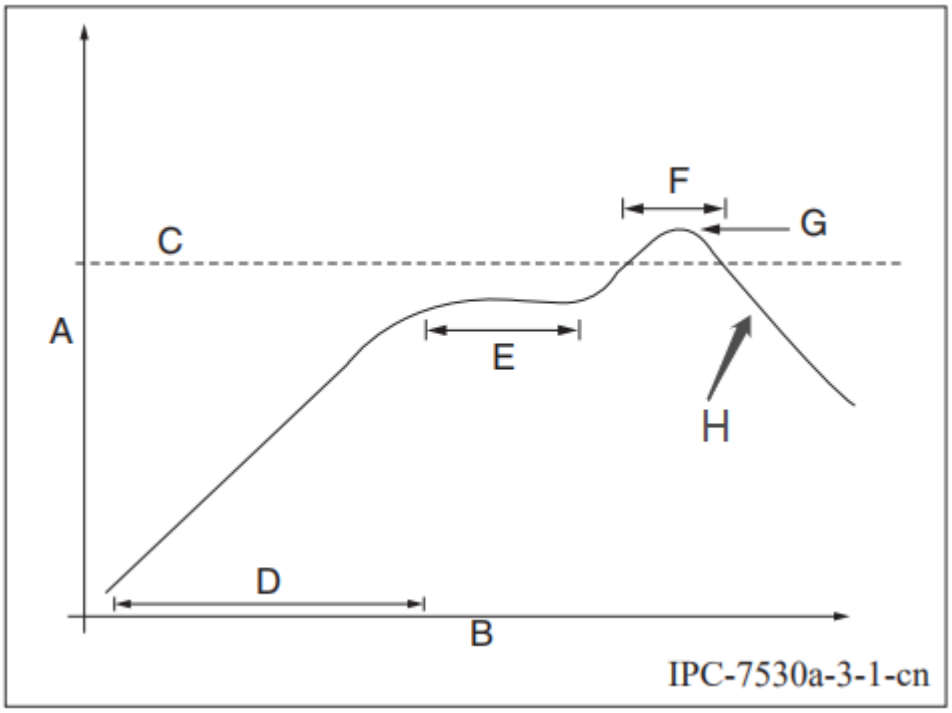
- Baking temperature: 40°C for reel packaging with relative humidity ≤ 5%. And 125°C for tray packaging with relative humidity ≤ 5% (use a heat-resistant tray, rather than plastic containers).

- Baking time: 168 hours for reel packaging and 12 hours for tray packaging.
- Temperature for triggering an alert: 50°C for reel packaging and 135°C for tray packaging.
- Production can begin after a module has cooled down to below 36°C under natural conditions.
- If a module remains unused for over 168 hours after being baked, it needs to be baked again.
- If a batch of modules is not baked after exposure for more than 168 hours, do not use wave soldering to solder them. Because these modules are level-3 moisture-sensitive devices, they are very likely to get damp when exposed beyond the allowable time. In this case, if they are soldered at high temperatures, device failure or poor soldering performance might occur.

5. In the whole production process, take electrostatic discharge (ESD) protective measures.
6. To guarantee the pass rate, we recommend that you use the SPI and AOI to monitor the quality of solder paste printing and mounting.

8.4. Recommended oven temperature curve

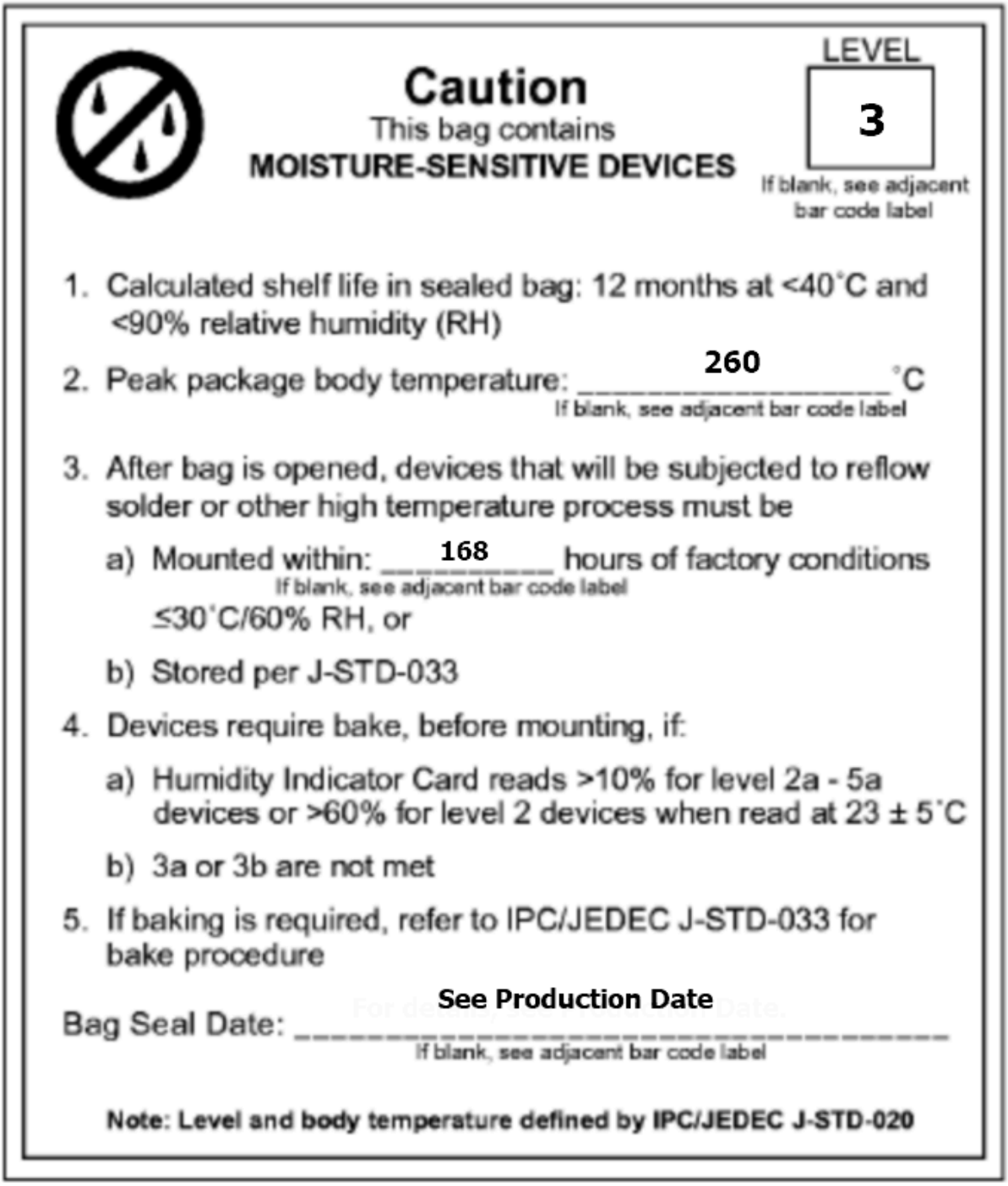
Set the temperature according to the following temperature curve of reflow soldering. The peak temperature is 245°C.



- A: temperature axis
- B: time axis
- C: alloy liquidus temperature from 217°C to 220°C
- D: ramp-up slope from 1°C/s to 3°C/s
- E: keep a constant temperature from 150°C to 200°C for a time period from 60s to 120s
- F: temperature above liquidus temperature for 50s to 70s
- G: peak temperature from 235°C to 245°C
- H: ramp-down slope from 1°C/s to 4°C/s

The curve above is based on solder paste SAC305. For more information about other solder pastes, see the recommended oven temperature curve in the specified solder paste specifications.

8.5. Storage conditions



Caution
This bag contains
MOISTURE-SENSITIVE DEVICES

LEVEL
3
If blank, see adjacent bar code label

1. Calculated shelf life in sealed bag: 12 months at <40˚C and <90% relative humidity (RH)
2. Peak package body temperature: 260 ˚C
   If blank, see adjacent bar code label
3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be
   a) Mounted within: 168 hours of factory conditions
      If blank, see adjacent bar code label
      ≤30˚C/60% RH, or
   b) Stored per J-STD-033
4. Devices require bake, before mounting, if:
   a) Humidity Indicator Card reads >10% for level 2a - 5a devices or >60% for level 2 devices when read at 23 ± 5˚C
   b) 3a or 3b are not met
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure

Bag Seal Date: See Production Date
If blank, see adjacent bar code label

Note: Level and body temperature defined by IPC/JEDEC J-STD-020

9. MOQ and packaging information

Product model	MOQ (pcs)	Shipping packaging	Modules per reel	Reels per carton
T5-E1-IPEX	3,600	Tape and reel	900	4

10. Appendix: Statement

FCC Caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user’s authority to operate this device. This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Note: This device has been tested and found to comply with the limits for a Class B digital device, according to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This device generates, uses, and can radiate radio frequency energy and, if not installed and used following the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this device does cause harmful interference to radio or television reception, which can be determined by turning the device off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the device and receiver.
- Connect the device into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Radiation Exposure Statement

This device complies with FCC radiation exposure limits set forth for an uncontrolled rolled environment. This device should be installed and operated with a minimum distance of 20cm between the radiator and your body.

Important Note

This radio module must not be installed to co-locate and operate simultaneously with other radios in the host system except by following FCC multi-transmitter product procedures. Additional testing and device authorization may be required to operate simultaneously with other radios.

The availability of some specific channels and/or operational frequency bands are country-dependent and are firmware programmed at the factory to match the intended destination. The firmware setting is not accessible to the end-user.

The host product manufacturer is responsible for compliance with any other FCC rules that apply to the host not covered by the modular transmitter grant of certification. The final host product still requires Part 15 Subpart B compliance testing with the modular transmitter installed.

The end-user manual shall include all required regulatory information/warnings as shown in this manual, including “This product must be installed and operated with a minimum distance of 20 cm between the radiator and user body”.

The RF module is considered as a limited modular transmitter according to FCC rules. Even though the RF module gets an FCC ID, the host product manufacturer can not use the FCC ID on the final product directly. In these circumstances, the host product manufacturer integrator will be responsible for re-evaluating the end product (including the transmitter) and obtaining the FCC authorization by a Class II permissive change application or a new application.

Declaration of Conformity European Notice



Hereby, Hangzhou Tuya Information Technology Co., Ltd declares that this module product is in compliance with essential requirements and other relevant provisions of Directive 2014/53/EU,2011/65/EU. A copy of the Declaration of Conformity can be found at https://www.tuya.com.



This product must not be disposed of as normal household waste, in accordance with the EU directive for waste electrical and electronic equipment

(WEEE-2012/19/EU). Instead, it should be disposed of by returning it to the point of sale, or to a municipal recycling collection point.

The device could be used with a separation distance of 20cm from the human body.